The Basel Problem

An impossible sum, a young genius, and a surprise visit from π

A slideSonnet demo

A simple question

What happens when you add the reciprocals of the perfect squares?

$$\frac{1}{1} + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \cdots$$

In modern notation:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = ?$$

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> *"If anyone finds and communicates to us that which thus far has eluded our efforts, great will be our gratitude."*

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$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

π? In a sum that has nothing to do with circles?

Step 1: start with $\sin x$

The Taylor series for sine:

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The coefficient of x^2 is $-\frac{1}{6}$ — hold onto it.

Step 2: factor by the roots

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$\frac{\sin x}{x}$ has zeros at $x = \pm\pi, \pm 2\pi, \pm 3\pi, \ldots$ Euler treated it like a polynomial and factored by its roots:

$$\frac{\sin x}{x} = \left(1 - \frac{x^2}{\pi^2}\right)\left(1 - \frac{x^2}{4\pi^2}\right)\left(1 - \frac{x^2}{9\pi^2}\right)\cdots$$

(Justified much later by the Weierstrass factorization theorem, 1876.)

Step 3a: reading the x^2 coefficient

With $a_k = \frac{1}{k^2\pi^2}$, three factors give

$$(1 - a_1x^2)(1 - a_2x^2)(1 - a_3x^2) = 1 - (a_1 + a_2 + a_3)\, x^2 + \cdots$$

Each x^2 term picks $-a_kx^2$ from *one* factor and 1 from the rest.

Step 3b: the infinite product

$$\prod_{n=1}^{\infty}\left(1-\frac{x^2}{n^2\pi^2}\right)=1-\left(\frac{1}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}\right)x^2+\cdots$$

So the x^2 coefficient is $-\dfrac{1}{\pi^2}\displaystyle\sum_{n=1}^{\infty}\frac{1}{n^2}$.

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From the Taylor series: $\dfrac{\sin x}{x} = 1 - \frac{1}{6}x^2 + \cdots$ From the product: $\dfrac{\sin x}{x} = 1 - \frac{1}{\pi^2}\sum \frac{1}{n^2}x^2 + \cdots$ Coefficients must agree:

$$\frac{1}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{1}{6}$$

The result

Multiply both sides by π^2:

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{\pi^2}{6} \approx 1.6449\ldots$$

The circle constant was hiding in the sum all along.

Why it matters

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Euler's proof was not rigorous by modern standards — but the answer was never in doubt. The sum is $\zeta(2)$, a value of the Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$. Even values are rational multiples of powers of π; the odd values remain mysterious.

What we learned

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{\pi^2}{6}$$

- A simple question can hide a profound answer.
- The boldness to treat $\sin x / x$ as a polynomial was the key.
- π connects geometry and number theory in ways we still explore.

Thanks for watching.